Controlled Evaluation of Class-Imbalance Mitigation Across Histopathology Patch Classification and WSI-Level MIL

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Abstract

This study evaluates methods for mitigating class imbalance in computational pathology, covering patch classification in CAMELYON16, TCGA-UT, BRACS, and PANDA and whole-slide multiple-instance learning (MIL) in CAMELYON16, TCGA-UT, and PANDA, all represented with frozen Virchow2 features. Patient-disjoint splits are created before imbalance is manipulated, and only training support changes: a full natural anchor is accompanied by size-matched balanced, moderate ($\rho = 10$), and severe ($\rho = 100$ where feasible) conditions. Validation and test patients, held-out prevalence, training-set size, model family, and numbers of optimizer updates remain fixed. Patch conditions retain a common independent-unit pool and manipulate patch allocation within it; MIL conditions report the independent slides and patients removed to create imbalance. The primary discrimination endpoint is balanced accuracy, evaluated alongside a co-primary tail-calibration endpoint so that imbalance harm to either axis qualifies a cell for mitigation analysis. The analysis first asks what predicts the existence and magnitude of the CE imbalance deficit across all cells, then evaluates mitigation recovery conditional on prespecified discrimination and calibration deficit gates. Secondary analyses address classwise discrimination, tail-sensitive calibration, computational cost, intrinsic separability, and condition-specific learnability.

Contents

1 Introduction

Class imbalance changes the empirical objective faced by a classifier: frequent classes supply more optimization terms, while rare classes may contribute too little evidence to establish stable decision boundaries. In pathology, the problem occurs at two distinct prediction units. Patch classification uses directly labelled crops, whereas WSI-level MIL predicts from bags whose instances inherit only a weak slide label. The same slide cohort can therefore be nearly balanced between diagnoses while containing very sparse class-informative tissue. Aggregate accuracy does not isolate either failure mode, motivating class-balanced discrimination and probability-quality evaluation (Saini and Susan, 2023; Zhang et al., 2023; Mosquera et al., 2024).

Native-dataset comparisons do not identify an imbalance effect by themselves. They simultaneously vary task, class separability, training support, label granularity, and evaluation prevalence. This study instead changes class allocation only in the training partition and retains identical natural validation and test cohorts. A size-matched balanced CE condition estimates attainable performance at the same training budget, so mitigation can be interpreted as recovery of damage attributable to the imposed imbalance rather than as an unconstrained gain over a convenient baseline.

The study addresses three research questions:

RQ1. Effect of imbalance. At fixed task, patient split, encoder, training-set size, and evaluation set, how does increasing training imbalance affect tail-class discrimination and calibration in patch classification and WSI-level MIL?

RQ2. Mitigation effectiveness. Which mitigation methods recover performance lost specifically to training imbalance, and what are their calibration and computational costs?

RQ3. Predictors of damage and mitigation headroom. Across all cells, do intrinsic class separability, condition-specific learnability, and independent support predict whether and how strongly imbalance damages CE performance? Conditional on demonstrated damage, do the same quantities predict mitigation recovery?

RQ1 establishes whether a recoverable problem exists. RQ2 conditions claims on that deficit. RQ3 separates an all-cell model of CE damage from a gate-conditional model of recovery and uses no test-derived predictor. Only TCGA-UT provides a genuine paired patch/WSI target; the other datasets either change the target between regimes or do not supply an unambiguous WSI label. All RQ3 regressions and regime interactions are therefore exploratory. The datasets and preprocessing provenance are documented in the companion dataset report, and all detailed method objectives and variants are defined in the companion methods report (Queisler, 2026a,b).

2 Experimental Design

The study varies training class support while holding the representation, prediction task, held-out cohorts, model family, and evaluation procedure fixed. The design comprises the datasets and prediction regimes, controlled data construction, a common mitigation roster, standardized training and selection, and explicit scope constraints.

2.1 Datasets and prediction regimes

Table 1 defines the prediction example and label in each available dataset–regime. A *patch-level example* is one eligible 256×256 crop and receives the label specified in the second column. A *bag-level example* is one WSI represented by its eligible frozen patch embeddings under the fixed instance cap and receives one label for the entire WSI. A patient or case is assigned to exactly one partition before any examples are selected. Where one patient contributes multiple slides, all slides follow the patient assignment. A fixed eligibility pipeline removes unusable samples before splitting and is identical across imbalance conditions.

Dataset	Patch-level example and label	MIL bag and bag label	Split unit
CAMELYON16	Non-overlapping tissue crop at 20x; tumor if at least 50% of its aligned mask cell is tumor, normal otherwise	One WSI bag; metastasis-bearing (tumor) versus normal slide	Slide (one slide per patient)
TCGA-UT	Author-supplied tumor-region crop; inherits its source WSI's cancer type	One diagnostic-WSI bag; the same one of 32 cancer types	TCGA participant
BRACS	Non-overlapping crop from a pathologist-annotated ROI; inherits one of seven ROI subtypes	Not defined or run: one WSI may contain ROIs with different subtypes and has no unique WSI label	Patient
PANDA	Foreground crop; cancer if at least 50% of its aligned provider-specific mask cell is cancer, benign otherwise	One biopsy-WSI bag; ordinal ISUP grade 0–5	Biopsy (one WSI)

Table 1: Dataset-specific definitions of patch-level examples and bag-level examples. Only TCGA-UT uses the same target in both regimes. CAMELYON16 and PANDA have different patch and bag targets, while BRACS has no unambiguous bag-level target.

Patch classification. Eligible patches are embedded once with frozen Virchow2. Concatenating the CLS token and mean patch token gives 2560 input features (Vorontsov et al., 2024; Zimmermann et al., 2024). For CAMELYON16 and PANDA, a patch label is derived from the spatially aligned expert mask and can differ from the WSI label. A TCGA-UT patch inherits the cancer-type label of the diagnostic WSI from which the dataset authors cropped it. A BRACS patch inherits the subtype of its pathologist-annotated ROI: normal (N), pathological benign (PB), usual ductal hyperplasia (UDH), flat epithelial atypia (FEA), atypical ductal hyperplasia (ADH), ductal carcinoma in situ (DCIS), or invasive carcinoma (IC). Patch eligibility, spatial sampling, label derivation, and the deterministic per-slide evidence cap are fixed before condition-specific training subsampling. The common classifier is a two-layer MLP with a 512-unit hidden representation, ReLU, dropout, and a linear class output, except where a method definition requires a different head.

WSI-level MIL. A slide is one bag of frozen patch embeddings; a patch is an instance, not a separately labelled training example in this regime. CAMELYON16 bags use the slide diagnosis (metastasis-bearing or normal), TCGA-UT bags use the diagnostic WSI's cancer type, and PANDA bags use the biopsy's ordinal ISUP grade from 0 to 5. BRACS is excluded from WSI-level MIL because its annotations label ROIs rather than whole slides and one slide can contain several ROI subtypes; assigning one of those subtypes to the complete slide would create an unsupported bag label. A common attention-MIL model maps instances to a 256-dimensional space, assigns normalized attention weights, and classifies the weighted bag representation. The per-slide instance cap and deterministic instance-selection seed are fixed across conditions. Slide labels are the only supervision; patch labels or masks are not used by the primary MIL training runs (Queisler, 2026a).

Comparative scope. Only TCGA-UT permits a direct patch-versus-WSI comparison. Both regimes predict the same cancer-type label within the same participant-disjoint cohort. In CAMELYON16, patch classification detects tumor tissue, whereas a WSI bag is classified as metastasis-bearing or normal. In PANDA, patch classification detects cancer tissue, whereas a WSI bag predicts the biopsy's ISUP grade. The CAMELYON16 and PANDA regimes are therefore separate dataset–target groups rather than paired measurements of one task. BRACS contributes only the seven-class ROI-patch task.

2.2 Controlled construction

For class supports $N_1, \ldots, N_K$, the headline imbalance ratio is

$$\rho = \frac{N_{\max}}{N_{\min}}. \tag{1}$$

The full distribution is also summarized by normalized entropy,

$$1 - H_{\text{norm}} = 1 - \frac{-\sum_{c=1}^{K} p_c \log p_c}{\log K}, \qquad p_c = \frac{N_c}{\sum_j N_j}. \tag{2}$$

The two quantities are complementary: ρ records the extreme head–tail contrast, whereas $1 - H_{\text{norm}}$ uses all classes and is normalized for class count. Both are computed separately at patch and slide level from each realized training manifest. Per-class patient, slide, and patch counts are reported alongside them.

Patient-disjoint training, natural-validation, and natural-test partitions are created first. Imbalance manipulation is confined to training support. Within a split, every condition uses exactly the same validation patients, test patients, and eligible held-out examples. No held-out example is deleted, duplicated, or reweighted according to the training severity. Patch-level evaluation uses every eligible held-out patch under the fixed evidence rule; slide-level evaluation uses every eligible held-out slide. Consequently, recall is estimated from all available held-out cases and prevalence-dependent precision and macro F1 are compared at one fixed test prevalence.

The validation cohort also retains its natural distribution. It is used for hyperparameter and checkpoint selection but never resized to resemble a training condition. Training, validation, and test manifests are versioned with patient identifiers, class counts, construction seed, and content hashes before model fitting.

Each eligible dataset–regime–split then has four training conditions:

1. the full natural training set, used only as a descriptive anchor;
2. a size-matched balanced training set;
3. a moderate imbalance condition targeting $\rho = 10$; and
4. a severe imbalance condition targeting $\rho = 100$ where feasible.

The balanced, moderate, and severe conditions contain the same total training support T: patches for patch classification and slides for MIL. The largest feasible shared T is chosen from training data alone under unique-support and method-feasibility constraints. The natural anchor may contain a different total and is excluded from the imbalance deficit and recovery estimands in Section 3.2.

The independent-evidence rule differs by regime. For patch classification, a fixed per-class patient pool and, within it, a fixed slide pool are shared by the balanced and all imbalanced conditions. Sampling first visits patients, then slides within patients, and only then patches. No patient may supply more than 10% of a class's patch allocation and no slide more than 5%; a deterministic round-robin pass through the contributing units enforces these caps before any unit is revisited. Thus patch imbalance changes patch allocation while preserving the available patient and slide diversity. For MIL, a slide is itself the labelled example, so changing slide support necessarily changes independent diversity. MIL selection first visits patients and then takes slide prefixes within them; each slide contributes once, and no patient supplies more than 10% of a class's slide allocation. Each manifest therefore reports, by class, unique patients, unique slides, patches, the largest patient and slide contribution, and the fraction of the eligible independent pool retained. RQ1 interprets MIL contrasts as the effect of controlled slide-support imbalance, including its induced change in patient diversity, rather than as a patch-count effect alone.

Minimum tail support is fixed from a training-only support-stability study conducted before definitive fitting. In the patch regime, an independent pilot unit is a patient, or a slide where patient and slide coincide. Every selected unit contributes a fixed quota q_{patch} of patches, apportioned as evenly as possible over its eligible slides. The quota is the largest common per-patient count feasible for every class at the largest available pilot level, subject to the contribution caps except for the documented five-patient pilot exception; it is determined from training inventories and then held constant across pilot support levels. In the MIL regime, the independent pilot unit is one slide represented by one capped bag, while patients contributing multiple slides remain identified for clustering.

Within each split and regime, nested balanced CE pilots use candidate independent-unit counts of $5, 10, 15, 20, 30$, or all available units when fewer exist. At the five-patient patch level, the 10% patient cap is arithmetically infeasible; the 5% slide cap can also be infeasible when the selected patients contribute too few slides. The pilot therefore selects five distinct patients with equal quotas and records this level-specific contribution-cap exception. At the five-slide MIL level, the 10% patient cap is likewise arithmetically infeasible: the pilot selects one slide from each of five distinct patients and records its level-specific exception. These five-unit pilots are used only for the support-stability curve, never as definitive training allocations. Three pilot construction seeds create three independent patient–slide–patch master orderings. Within an ordering, every candidate is a prefix of the next candidate and uses the same held-out cohort, so support-level contrasts are paired and nested. The *stability floor* is the smallest count for which the increase at the next candidate is below 0.01 in mean natural-validation balanced accuracy and below 0.02 for every class across the three pilot orderings; if no candidate satisfies this rule, the largest candidate is used. The definitive minimum for every class is the maximum of this stability floor and the method floor. The method floor is two independent same-class examples per set or batch and at least 10 distinct contributing patients and 20 slides for patch training, or 10 patients and 20 slides for MIL, except where patient and slide coincide, in which case 20 slides are required. Every class-aware batch is constructed with at least two distinct same-class independent units. These are eligibility constraints, not examples created by replacement. A requested severity is lowered to the largest allocation satisfying both floors; a dataset–regime is excluded rather than run as a few-example experiment if even the balanced condition fails them. Pilot quotas, results, seeds, and resulting floors are frozen before definitive training and reported with the manifests.

For a K-class exponential construction, an assigned class order receives weights

$$w_i(\rho) = \rho^{-i/(K-1)}, \qquad n_i(\rho) \approx T \frac{w_i(\rho)}{\sum_{j=0}^{K-1} w_j(\rho)}, \quad i = 0, \ldots, K-1. \tag{3}$$

Constrained integer allocation preserves $\sum_i n_i = T$, satisfies the independent-support floors, and never exceeds available unique training examples. Binary targets use $n_{\mathrm{head}}/n_{\mathrm{tail}} = \rho$ with the same total T. A prespecified step construction may be used for clinically meaningful BRACS groups; its class grouping and realized supports are reported before analysis. One master random ordering is created per class at each hierarchy level: patients, slides within patients, and patches within slides. Every condition is a deterministic prefix or round-robin prefix of these same orderings. For patch classification, all fixed-pool patients and slides contribute before additional patches are allocated; for MIL, class-specific slide selection is a patient-round-robin prefix of the same hierarchical ordering.

If a target ratio is infeasible, the construction uses the largest attainable ratio and reports the requested ratio, achieved ratio, limiting class, realized counts, and binding independent-support floor. It never reduces validation or test support to manufacture feasibility. In particular, TCGA-UT patch classification uses balanced, $\rho = 10$, and its maximum feasible or native-severity condition instead of forcing $\rho = 100$. A stable lower-severity condition is retained in preference to a nominal $\rho = 100$ condition supported by only a few independent units.

Class identity must not be confounded with tail status. Multiclass targets therefore use three locked tail assignments where feasible: (i) a clinical or native-support assignment, (ii) a reversed assignment for ordinal targets or a deterministic rotation for nominal targets, and (iii) one random class permutation drawn once and recorded before fitting. Binary tasks use both head/tail orientations. If support constraints make an assignment infeasible at the shared T, the achieved allocation is reported and the assignment is omitted only under a prespecified feasibility rule.

Five seed families remain independent. Patient-split seeds determine cohort membership; assignment seeds determine the locked class order; construction seeds determine the master ordering and evidence selection within patients and slides; initialization seeds determine optimization randomness; and bootstrap or permutation seeds determine resampling. The three construction seeds used by the stability pilot are disjoint from the fourth construction seed used to create definitive manifests. Two locked tuning initialization seeds are used for every hyperparameter candidate within each split and assignment. They are disjoint from the five confirmation initialization seeds used after selection. No seed is chosen based on test performance.

The size-matched balanced manifest and its trained CE baseline are identical for all tail assignments within a split: assignment is undefined at $\rho = 1$. The same balanced predictions enter every assignment-specific deficit and are represented once in resampling data, avoiding duplicated balanced fits, bootstrap records, and artificial assignment variability.

2.3 Mitigation methods

Table 2 is intentionally compact. Equations, assumptions, defaults, and fidelity qualifications reside in the methods report (Queisler, 2026b). CE and MIL CE are the size-matched no-mitigation references. Balanced Softmax is not a separate tested method: it coincides with the $\tau = 1$ special case of the train-time logit adjustment already in the roster, and is cited only as related prior-correction literature (Ren et al., 2020).

Family	Tested methods	Regime
Reference	CE; MIL CE	Patch; WSI
Sampling	Balanced sampling	Both
Loss and prior correction	Weighted CE; focal loss; train-time and post-hoc logit adjustment; CFAL	Both, except CFAL patch only
Hybrid objective	CE + soft F1; CE + soft MCC; OKO set learning	Patch
Feature mixing	RankMix-inspired feature mixing	WSI
Representation/classifier	cRT; SC-MIL	Both for cRT; WSI for SC-MIL
Dual expert	MDE-inspired dual-expert MIL, including no-consistency ablation	WSI

Table 2: Prespecified method roster. “Both” means the patch and WSI implementations defined in the companion methods report, not a comparison of their raw scores.

RankMix-inspired denotes the tested adaptation with teacher ranking, representative subsequences, feature interpolation, mixed labels, and a reinitialized soft-label-CE student, but without the source paper’s complete combined instance and bag objectives (Chen and Lu, 2023). MDE-inspired denotes the dual-expert component under frozen Virchow2 bags without multimodal distillation; $\lambda_{\text{con}} = 0$ is the dual-expert-without-consistency ablation (Ling et al., 2025). SC-MIL uses class-aware batches and records valid anchors and positive-pair counts. Logit adjustment uses training-condition priors only (Menon et al., 2021); cRT reinitializes and retrains only the final classifier after freezing the learned patch representation or WSI instance encoder and attention aggregator (Kang et al., 2020).

2.4 Training, selection, and replication

The frozen encoder, classifier family, optimizer family, batch definition, augmentation, and numerical precision are held fixed within a regime. Training is budgeted only in optimizer updates; epochs are neither a stopping rule nor a unit of comparison. Let B be the regime's locked batch size and T the common controlled support. A single-stage model receives

$$U = 30 \left\lceil \frac{T}{B} \right\rceil \tag{4}$$

updates, corresponding to 30 reference passes through T examples under ordinary sampling. Balanced sampling, replacement, synthetic bags, and dual loaders do not change U. RankMix receives U teacher updates and, after reinitialization, U student updates. cRT receives U stage-one updates and $\lceil 0.2U \rceil$ stage-two classifier updates. MDE performs U joint updates, each consuming its prespecified natural and balanced minibatches. The natural anchor uses the same formula with its own support and is descriptive. Processed examples, unique-example exposures, and updates are all reported. Each run stores validation metrics at fixed update intervals. The selected checkpoint maximizes natural-validation balanced accuracy, with macro F1 and then lower NLL as deterministic tie-breakers. Test predictions are produced once from that locked checkpoint.

Hyperparameter comparisons are bounded by a common validation-search budget rather than by restricting every method to one tuned quantity. For each trainable mitigation method, its four-value imbalance-specific grid is crossed with the common four-value learning-rate grid in Appendix A, yielding at most 16 candidate configurations. This joint search accommodates interactions between a method's objective and optimization dynamics while holding the optimizer family, weight decay, architecture, batch definition, and all other training parameters fixed. CE and methods without a meaningful imbalance-specific control evaluate the four learning rates without introducing an artificial search dimension. Train-time logit adjustment is tuned jointly over τ and learning rate; post-hoc logit adjustment inherits the selected CE model and tunes only τ because it performs no gradient optimization.

Every hyperparameter candidate is fitted with the same two tuning initialization seeds for each prespecified patient split and tail assignment. One configuration is selected per method, dataset, prediction regime, and imbalance severity by aggregate natural-validation balanced accuracy across both tuning seeds, the tuning splits, and the assignments. Macro F1 and then lower NLL are deterministic tie-breakers. The chosen configuration is locked before any of the five disjoint confirmation initialization seeds are fitted and before test evaluation. Tuning runs never enter confirmation effect estimates. For cRT, stage one inherits the selected CE configuration, while the stage-two classifier learning rate is selected separately from the common grid. Temperature scaling, when reported, fits one positive scalar on natural-validation NLL after model and checkpoint selection and applies it unchanged to test logits (Guo et al., 2017).

Controlled contrasts are paired on split, assignment, confirmation initialization, and held-out cases. Each locked patient split and tail assignment is repeated with the five confirmation seeds. Exactly three patient-split seeds are used; a dataset–regime unable to produce all three valid splits under the independent-support floors is excluded from the corresponding confirmatory analysis. The split counts, support floors, condition manifests, tuning and confirmation seeds, method grids, and update budgets are frozen in the signed analysis manifest before the first definitive fit. The unit of replication and all missing or failed runs are reported. A failed run is not silently replaced using a test-informed seed.

Computational cost is reported separately for validation search and locked-configuration fitting, using wall-clock training time, accelerator-hours, peak accelerator memory, total and trainable parameter counts, and effective passes through unique training examples. Costs include all stages required by the method, including both cRT stages, but exclude one-time frozen-feature extraction shared by all methods. Hardware and software environments are recorded with each run.

2.5 Scope and limitations

The design isolates training-distribution effects under frozen Virchow2 representations; it does not establish how end-to-end encoder fine-tuning would respond to imbalance. Constructed tails improve causal interpretability but need not reproduce referral pathways or biological prevalence. A shared T controls data volume at the cost of discarding some available training examples, which is why the full natural condition is retained as a descriptive anchor. Some methods change more than one mechanism, so the study supports method-level comparisons rather than universal causal claims about sampling or loss design. The locked update budget equalizes optimizer steps but not information exposure: RankMix consumes teacher and student passes, MDE draws a natural and a balanced minibatch per update, and post-hoc logit adjustment performs no gradient optimization at all, so method comparisons are made at equal update budget with exposure and cost reported alongside, not at equal data throughput. The number of independent patients, not the number of patches, limits inference. All imbalance-robustness findings are additionally conditional on the geometry of a single frozen encoder; whether the same deficits and recoveries hold under a different foundation representation is outside the present scope. Finally, calibration on a natural held-out cohort is conditional on that cohort's prevalence and should not be transported to a new deployment population without explicit prior-shift analysis.

3 Evaluation and Statistical Analysis

Evaluation separates predictive discrimination, probability quality, mitigation recovery, and exploratory predictors of damage and recovery. All estimands and inferential families are defined before test evaluation.

3.1 Endpoints and aggregation units

The primary endpoint is balanced accuracy, equal to macro recall for single-label multiclass prediction. Secondary discrimination outcomes are macro F1; per-class recall and F1; and head, body, and tail recall under the locked assignment. Classes are ranked by allocated training support: the first $\lceil K/3 \rceil$ ranks are head, the last $\lceil K/3 \rceil$ are tail, and intervening ranks are body; binary tasks have one head and one tail and no body. Ties follow the locked class ordering. Quadratic-weighted kappa and ordinal mean absolute error are additional secondary outcomes for PANDA ISUP grade. Overall accuracy is descriptive. Patch predictions are summarized in two ways: patch-micro estimates treat eligible patches as predictions while uncertainty remains patient-clustered, and slide/patient-macro estimates first aggregate the metric contribution within each slide or patient so that heavily tiled slides do not dominate. WSI outcomes are slide-level with patient clustering where patients contribute multiple slides.

Let N denote the number of held-out predictions in an aggregation unit, C the number of classes, $y_i \in \{1, \ldots, C\}$ the true label, and $\hat{\mathbf{p}}_i \in [0, 1]^C$ the predicted probability vector. Write Y_i for the one-hot encoding of y_i. Probability quality is assessed at natural-prevalence and class-balanced weightings.

Negative log likelihood. Natural-prevalence negative log likelihood (NLL), also called multiclass log loss, is

$$\mathrm{NLL} = -\frac{1}{N}\sum_{i=1}^{N} \log \hat{p}_{i,y_i}. \tag{5}$$

Lower NLL indicates that more probability mass is assigned to the true class. The implementation clips $\hat{p}_{i,y_i}$ to $[10^{-12}, 1]$ before taking the logarithm. To prevent common classes from dominating probability-quality summaries, class-conditional NLL and macro NLL are

$$\mathrm{NLL}_c = -\frac{1}{N_c}\sum_{i:y_i=c} \log \hat{p}_{ic}, \qquad \mathrm{NLL}_{\mathrm{macro}} = \frac{1}{C}\sum_{c=1}^{C} \mathrm{NLL}_c, \tag{6}$$

where N_c is the number of held-out predictions from class c. Natural-prevalence NLL is the deployment-facing summary for the observed held-out cohort; macro NLL and unweighted head, body, and tail averages of NLL_c are tail-sensitive summaries.

Multiclass Brier score. The multiclass Brier score penalizes squared error across the full predicted distribution (Brier, 1950):

$$\mathrm{BS} = \frac{1}{N}\sum_{i=1}^{N} \|\hat{\mathbf{p}}_i - Y_i\|_2^2. \tag{7}$$

Lower values indicate better probability quality. The natural-prevalence score averages this error over the complete held-out cohort. Classwise scores apply the same full-distribution error only to predictions with $y_i = c$. Their unweighted averages provide head, body, and tail summaries.

Expected calibration error. For each prediction, let $\hat{y}_i = \arg\max_c \hat{p}_{ic}$ and $q_i = \max_c \hat{p}_{ic}$. Expected calibration error (ECE) partitions confidence into $B = 10$ fixed equal-width bins on $[0, 1]$ and compares mean confidence with empirical accuracy in each nonempty bin (Naeini et al., 2015):

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{|I_b|}{N} \left| \frac{1}{|I_b|}\sum_{i\in I_b} \mathbb{1}[\hat{y}_i = y_i] - \frac{1}{|I_b|}\sum_{i\in I_b} q_i \right|, \tag{8}$$

where I_b contains the predictions assigned to bin b. ECE is secondary because it depends on binning and can be unstable with small classwise samples; its bootstrap uncertainty uses the same fixed bins and patient weights as the other endpoints. Reliability diagrams show the binwise confidence–accuracy relation overall and for the tail group.

Because checkpoints and hyperparameters are selected for natural-validation balanced accuracy, all probability-quality results describe discrimination-selected models rather than models optimized for calibration. Temperature scaling and target-prior correction are the only calibration-specific adjustments applied after selection.

Scores for balanced decisions are kept distinct from probabilities for the natural target prevalence. For an ordinary CE score z_c^{CE} and realized training prior π_c^{train}, post-hoc logit adjustment uses $z_c^{\mathrm{CE}} - \tau \log \pi_c^{\mathrm{train}}$ for balanced-accuracy decisions. It is not reported as a naturally calibrated probability. Under the prior-shift assumption, target-prevalence probabilities instead normalize

$$z_c^{\mathrm{CE}} - \log \pi_c^{\mathrm{train}} + \log \pi_c^{\mathrm{target}}. \tag{9}$$

For train-time logit adjustment, training applies CE to $g_c = z_c + \tau \log \pi_c^{\mathrm{train}}$. Because g_c approximates training-posterior logits, the corresponding target-prevalence logits are

$$z_c + (\tau - 1)\log \pi_c^{\mathrm{train}} + \log \pi_c^{\mathrm{target}}. \tag{10}$$

The unadjusted z_c remains the train-time method's validation-selected discrimination score but is a fully prior-stripped score only when $\tau = 1$. The target prior π^{target} is estimated separately within each natural-validation split and applied only to the corresponding test split. Temperature scaling is fitted on natural-validation NLL after the appropriate target-prior correction and applied unchanged to test logits. Raw, balanced-decision, target-prior-corrected, and temperature-scaled outputs are retained separately. A single temperature is interpreted as a sharpness correction, not a remedy for class-specific prior mismatch.

3.2 Imbalance deficit, recovery, and inference

For a higher-is-better metric M, define the imbalance-induced deficit and recovery fraction at each imbalanced condition as

$$\begin{aligned} D_M &= M(\text{balanced CE}) - M(\text{imbalanced CE}), \\ R_M &= \frac{M(\text{method on imbalanced data}) - M(\text{imbalanced CE})}{D_M}. \end{aligned} \tag{11}$$

$R_M = 0$ denotes no recovery, $R_M = 1$ recovery to the size-matched balanced CE reference, $R_M < 0$ further deterioration, and $R_M > 1$ performance beyond that reference. Because the balanced reference draws more unique tail independent units than any imbalanced condition at the same T, methods that only reweight or recalibrate the imbalanced manifest cannot restore tail evidence that was never sampled; for that family $R_M < 1$ may reflect an information ceiling rather than method failure, whereas resampling and synthesis methods are the only family for which $R_M = 1$ is mechanically attainable. The full natural condition is descriptive and never enters D_M or R_M.

Mitigation effectiveness is evaluated through two prespecified co-primary gates, so that imbalance harm to either discrimination or tail probability quality qualifies a cell. The *discrimination gate* opens a dataset–regime–severity–assignment cell when the CE balanced-accuracy deficit is at least 0.02 and its paired 95% confidence interval excludes zero. The *calibration gate* opens the same cell when the CE tail-group macro-NLL deficit, $\text{NLL}_{\text{tail}}(\text{imbalanced CE}) - \text{NLL}_{\text{tail}}(\text{balanced CE})$, is at least 0.05 nats with a paired 95% confidence interval excluding zero. Both gates are computed from CE runs on the same held-out cases before any mitigation method is compared. A cell entering through the discrimination gate is evaluated for balanced-accuracy recovery; a cell entering through the calibration gate is evaluated for tail-calibration recovery; a cell may enter through both. Separating the gates prevents the common regime in which strong frozen features leave balanced accuracy nearly intact while imbalance still biases tail probabilities, which a single balanced-accuracy gate would silently exclude even though prior-correction methods are expected to act there. Both gates prevent unstable recovery ratios when there is no evidenced damage to recover, and all CE deficits on both axes are reported, including gated-out cells. For the lower-is-better calibration axis the recovery fraction uses the sign-corrected deficit $D_{\text{cal}} = \text{NLL}_{\text{tail}}(\text{imbalanced CE}) - \text{NLL}_{\text{tail}}(\text{balanced CE})$ as denominator and the method's tail-NLL improvement over imbalanced CE as numerator, so the interpretation of Equation (11) carries over. Recovery for other secondary metrics remains descriptive and does not reopen a cell that fails both gates.

Confidence intervals use a class-preserving crossed paired bootstrap. Before model training, 10,000 label-only replicates are generated from the locked test manifests. Each patient is assigned a stratum defined by its complete split-by-class contribution vector across all test appearances; this vector can contain several classes for patch targets. Within every nonempty stratum, the observed number of patients is resampled with replacement. A selected patient's multiplicity is then applied to all of that patient's slides and patches in every split, so overlapping test appearances never become independent people. When patient and slide coincide, slides are the resampling units. Preserving every observed contribution stratum guarantees that each class represented in an original split remains represented in every replicate; an uncomputable split-level metric is treated as an implementation failure and the analysis stops rather than silently discarding the replicate.

The label-only preflight verifies class representation, computability of every split-level metric, and identical patient multiplicities across split appearances. It also records, by split and class, the number of unique resampled patients, Kish effective patient count, and largest normalized patient weight. If the Kish count falls below five in any replicate, or one patient supplies more than 50% of a class's bootstrap weight in more than 5% of replicates, inference for that dataset–regime is designated descriptive before training. The diagnostic report and resampling seed are frozen with the analysis manifest.

In the model-based bootstrap, the three split seeds are fixed design repetitions and are averaged with equal weight, not resampled as independent cohorts. Tail assignments remain separate unless an explicitly assignment-averaged estimand is reported. Within each replicate, the five matched confirmation initialization indices are resampled as paired algorithmic-noise blocks and averaged before split aggregation; they never increase the patient sample size. Balanced CE, imbalanced CE, and each method use the same patient and initialization draws. The shared balanced baseline is loaded once and receives the same bootstrap weights in every assignment-specific deficit. The full deficit and recovery ratio in Equation (11), including its numerator and denominator, is recomputed inside every replicate. Patches are never bootstrapped independently. Percentile 95% intervals are reported with the number of replicates and seed fixed in the analysis manifest.

The tested effect is gate-specific. On the discrimination axis it is the raw balanced-accuracy difference $\Delta_{\mathrm{BA}} = \mathrm{BA}(\text{method}) - \mathrm{BA}(\text{imbalanced CE})$; on the calibration axis it is the paired tail-group NLL change $\Delta_{\mathrm{cal}} = \mathrm{NLL}_{\mathrm{tail}}(\text{imbalanced CE}) - \mathrm{NLL}_{\mathrm{tail}}(\text{method})$, oriented so positive values indicate improvement. Recovery ratios are reported as normalized effects and are never test statistics. Raw two-sided p-values use a paired patient-block permutation test. Under each permutation, the complete method and CE prediction blocks are swapped within a patient using the same swap for all slides, patches, split appearances, and matched confirmation seeds, after which the gate's statistic is recomputed. All permutations are enumerated when feasible; otherwise 100,000 random permutations and the plus-one correction are used.

To keep the confirmatory family commensurate with the available independent-patient evidence, a prespecified *primary* method set is confirmatory and the remaining roster is exploratory. The primary set contains the loss-reweighting, prior-correction, and resampling references that directly target tail support and are shared across both regimes: weighted CE, focal loss, train-time logit adjustment, and balanced sampling. The confirmatory family for Holm adjustment is every primary-method $\times$ gate $\times$ severity $\times$ tail-assignment comparison within a dataset and regime; severities are not separate families, and each co-primary gate contributes its own statistic. Holm's procedure adjusts the permutation p-values across that family, with gated-out comparisons recorded as not tested. All other methods (hybrid objectives, CFAL, OKO, cRT, RankMix-inspired, SC-MIL, MDE-inspired, and post-hoc logit adjustment) are reported with the same effect estimates and intervals but form a separate exploratory family outside the confirmatory correction. Effect estimates and intervals remain primary throughout; adjusted p-values are supporting evidence. Secondary classwise, calibration, and cost analyses are exploratory unless explicitly prespecified above.

3.3 Predictors of damage and mitigation headroom

Two distinct frozen-feature measurements are made before mitigation fitting. *Intrinsic separability* is measured once per dataset, regime, and split using a fixed balanced reference subset drawn from the eligible training pool and shared by every condition. *Condition-specific learnability* uses the actual controlled training manifest as its neighbour reference or probe-fitting set. Both use (i) balanced k-nearest-neighbour macro recall, (ii) a class-balanced linear probe, and (iii) per-class nearest-neighbour error, measured on the unchanged natural-validation set.

They never use test labels or CE test performance. Patch probes operate on patch embeddings. WSI probes use the mean and componentwise standard deviation of the capped instances, fixed non-learned summaries that do not inherit the attention-MIL model being explained.

Effective sample size accounts for clustered patch evidence:

$$N_{\mathrm{eff},c} = \frac{N_c}{1 + (\overline{m}_c - 1)\,\mathrm{ICC}_c}, \tag{12}$$

where $\overline{m}_c$ is mean cluster size and ICC_c is the within-patient or within-slide intraclass correlation of a scalar class margin. That margin is the fixed intrinsic linear probe's logit for class c minus its largest competing logit, evaluated on training-pool embeddings by cross-fitting so an observation is not scored by a probe fitted on itself. Unique patient and slide counts are the primary support measures. Equation (12) is a secondary sensitivity measure for patch targets, not a replacement for independent-unit counts. Slide count itself is used for WSI targets, with patient count also entered when patients contribute multiple slides.

RQ3 uses three linked exploratory analyses rather than a hurdle model for the signed deficit. First, a logistic model describes whether a cell passes either prespecified deficit gate in Section 3.2. Second, a continuous model uses D_{BA} from every cell, including negative deficits. Its observation model incorporates the bootstrap standard error $s_{D,j}$, so $\widehat{D}_j$ is modeled with variance $s_{D,j}^2 + \sigma^2$ rather than treating estimated deficits as error-free. Third, recovery is modeled only among gated cells and is explicitly interpreted as conditional on demonstrated damage.

Because the independent unit for RQ3 is the dataset–target group and only a few exist, the analyses are deliberately restricted rather than predictive. Each model uses a minimal two-predictor set chosen from a prior hypothesis: $\log \rho$ and intrinsic separability, both standardized. Condition-specific learnability, log minimum independent support (patients for patch targets and slides for MIL targets), and the patch-versus-WSI indicator are reported as descriptive covariates and entered only in prespecified single-predictor sensitivity fits, never jointly, because with so few groups they are collinear with the two primary predictors and with each other. Dataset–target groups receive random intercepts rather than fixed dummy effects; the recovery model uses the same two predictors and deliberately omits per-method effects, which the available gated cells cannot identify. Zero-centered unit-scale Gaussian priors on standardized slopes and on the log standard deviations of the random intercept and residual terms provide proper MAP regularization, and no interactions are fitted. Leave-one-dataset–target-group-out cross-validation is reported only as a coarse stability check: with roughly four groups its held-out predictions collapse toward the marginalized population intercept, so RQ3 is interpreted as hypothesis-generating description of association rather than a validated predictive model. TCGA-UT supplies the only genuine paired-regime evidence; the CAMELYON16 and PANDA targets differ by regime, and BRACS has no WSI regime. Model definitions, priors, and covariate roles are frozen before recovery outcomes are inspected, and all RQ3 results remain exploratory.

A Experimental Controls

Every trainable method uses the common learning-rate grid

$$\eta \in \{1 \times 10^{-4}, 3 \times 10^{-4}, 1 \times 10^{-3}, 3 \times 10^{-3}\}. \tag{13}$$

For methods in Table 3, this grid is crossed with the four candidate values of the method-specific control. The resulting maximum is 16 configurations per method, dataset, regime, and severity. Any computationally necessary narrowing is completed and recorded in the signed analysis manifest before the first definitive fit; no grid is changed once definitive training begins.

Method or family	Candidate method-specific control
Balanced sampling / weighted CE	strength $\in \{0.25, 0.5, 0.75, 1\}$
Focal loss	$\gamma \in \{0.5, 1, 1.5, 2\}$
Logit adjustment	$\tau \in \{0.25, 0.5, 1, 2\}$
CE + soft F1 / soft MCC	auxiliary weight $\in \{0.25, 1, 4, 16\}$
CFAL	affinity bandwidth $\sigma \in \{0.25, 1, 2, 4\}$
OKO	odd-class count $k \in \{1, 2, 4, 8\}$, capped by $K - 1$
RankMix-inspired	beta shape $\alpha \in \{0.5, 1, 2, 4\}$
SC-MIL	temperature $\tau \in \{0.05, 0.1, 0.2, 0.5\}$
MDE-inspired	consistency weight $\lambda_{\text{con}} \in \{0, 0.1, 0.25, 0.5\}$

Table 3: Prespecified method-specific grids crossed with the common learning-rate grid in Equation (13). cRT has no imbalance-strength grid; only its stage-two learning rate is selected separately.

For every realized condition, the experiment record contains dataset version; eligibility rules; patient, slide, and patch identifiers by partition; requested and achieved T and ρ; class assignment; all seed families and their tuning, confirmation, pilot, or definitive roles; pilot patch quota; evidence caps; bootstrap-preflight diagnostics; model and optimizer configuration; candidate grid; selected checkpoint; environment identifier; and test-prediction hash. Deviations are dated and justified before the affected test labels are accessed.

B Terminology

Natural anchor. The full eligible natural training partition. It is descriptive and may have a different size from controlled conditions.

Size-matched balanced reference. A training subset with total support T and approximately equal class counts; it defines the counterfactual reference in D_M.

Imbalance severity. The requested training allocation summarized by ρ; achieved severity is computed from realized integer counts.

Tail assignment. The locked mapping from semantic classes to positions in the constructed support profile.

Imbalance deficit. The paired balanced-CE minus imbalanced-CE difference at common T and on the same held-out cases.

Recovery. A method's improvement over imbalanced CE divided by the evidenced imbalance deficit.

Patch-micro estimate. A metric computed over eligible patch predictions, with patient-clustered uncertainty.

Patient-macro estimate. An equal-weight aggregation over slides or patients, limiting domination by heavily tiled cases.

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